# Electro-hydraulic command-to-motion chain

**ELECTRICAL**

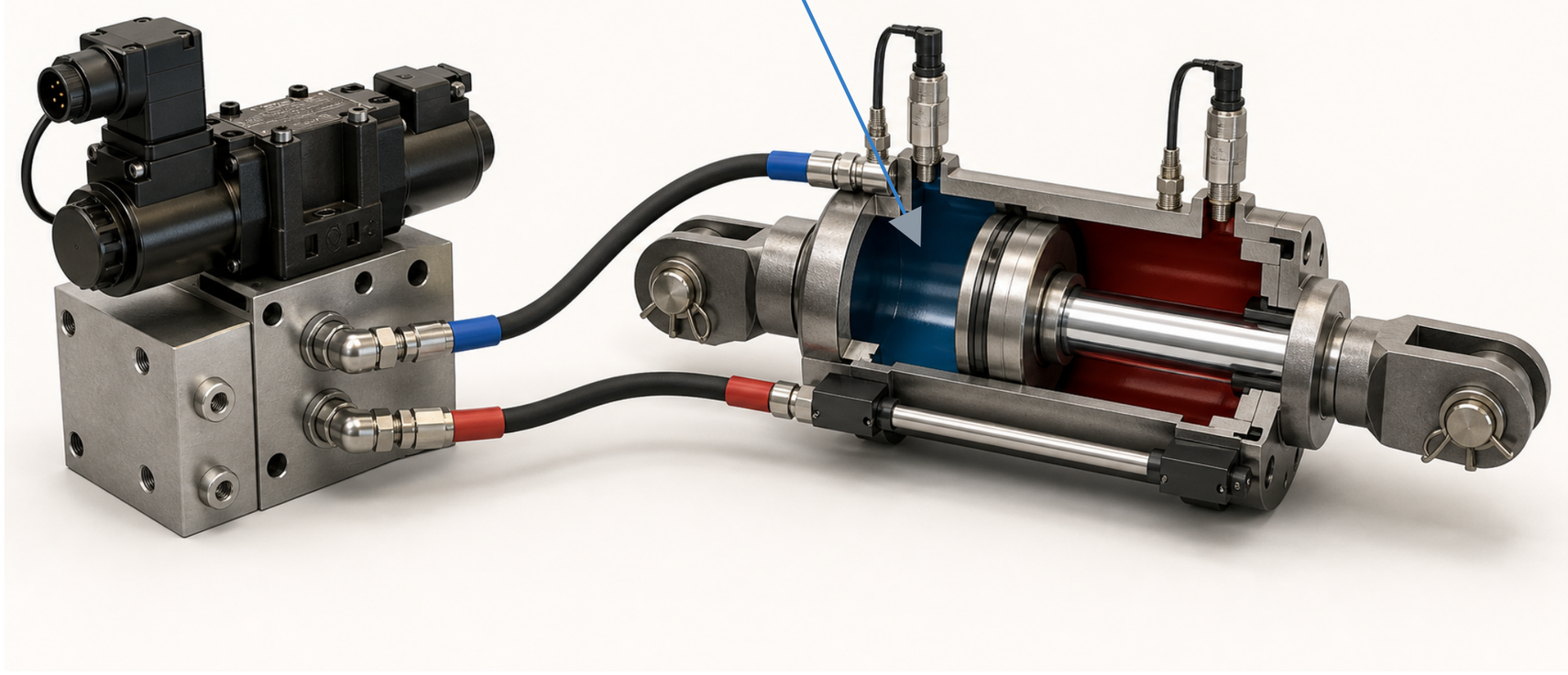
$u \rightarrow$ spool $x_v$

**HYDRAULIC**

$q \rightarrow p_1, p_2$

**MECHANICAL**

$F_h \rightarrow x, \dot{x}$



**VALVE**

dead zone · overlap · saturation

**FLUID**

orifice law · compressibility · leakage

**LOAD**

piston friction · inertia · external force

load pressure changes the valve pressure drop

piston motion consumes chamber flow